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Research article

Quantification of Pigment Epithelium-Derived Factor (PEDF) in an Ex vivo Coculture of Retinal Pigment Epithelium Cells and Neuroretina

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Abstract

To quantify the pigment epithelium-derived factor (PEDF) levels in a coculture model of physically separated neuroretina and retinal pigment epithelium (RPE) cells, and to point out its potential role in neuroretinal maintenance. RPE cells and neuroretina explants were isolated from porcine eyes. RPE cells were expanded and seeded on the bottom of Transwell® culture inserts. Neuroretina explants were cultured alone (controls) on Transwell® culture membranes or supplemented with RPE cells in the same wells but physically separated by the culture membrane, during 9 days. PEDF concentration in the culture medium at 3, 5, 7 and 9 days of culture was determined by enzyme-linked immunosorbent assays (ELISA) specific for porcine samples. Mean statistical analysis were performed with Pair-wise Student's-tests.bCulture medium collected from neuroretina cultures without and with RPE cells contained detectable levels of PEDF in both conditions and at all evaluated time-points. At 3, 5, and 7 days and through the whole time of culture PEDF concentration was significantly higher in cocultures with RPE cells ($p<0.05$). RPE cells neuroprotective role may be linked to the beneficial effects of neurotrophic factors, such as PEDF, secreted or induced by RPE cells during co-culture.

Keywords

Ex vivo model; Retinal pigment epithelium (RPE); Neuroretina; Pigment Epithelium- Derived Factor (PEDF); Neuroprotection

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Introduction

Neuroretinal cultures have been commonly used to advance in the knowledge of retinal physiology and pathobiology, as well as to evaluate potential therapies for retinal degeneration diseases (Alt et al., 2013; Ghosh et al., 2012; Martínez-Fernández de la Cámara et al., 2013). Our group has routinely used neuroretina explant cultures during the last 15 years to evaluate retinal dynamics under different in vitro conditions. Thus, in this ex vivo model our studies have confirmed the implication of monocytes/macrophages and/or pro-inflammatory cytokines, such as tumor necrosis factor (TNF) alpha, in the pathobiology of the proliferative vitreoretinopathy (PVR) (Fernandez-Bueno et al., 2013, 2008). Furthermore, it has been explored the role of anti-inflammatory agents such as Adalimumab (Humira®, Abbott Laboratories Ltd.), a TNF-blocker, for the reduction of reactive gliosis during retinal degeneration process in vitro (Fernandez-Bueno et al., 2013). We have also used the neuroretina cultures to evaluate the potential of mesenchymal stem cells (MSC) therapy for retinal neuroprotection. It has been established a mixed co-culture of three cellular layers, neuroretina, retinal pigment epithelium (RPE) cells, and MSC (Rodriguez-Crespo et al., 2014). And we reported that MSC could have paracrine neuroprotective effects for the neuroretina, through elevated concentrations of brain-derived neurotrophic factor and ciliary neurotrophic factor (Labrador-Velandia et al., 2019). Finally, we have recently used this ex vivo model to determine the neuroretinal cytotoxicity potential of intraocular endotamponades (Pastor et al., 2017).

During this time, we have also developed a coculture model of neuroretina supplemented with physically-separated RPE cells to mimic ex vivo the subretinal space condition that develops during retinal detachment pathology (Di Lauro et al., 2016), where the neuroretina and the RPE become physically separated. In normal retina, the RPE plays a key role for photoreceptors homeostasis and functioning, performing the synthesis and maintenance of the interphotoreceptor matrix, photoreceptor membrane turnover, and retinoid metabolism (Strauss, 2005). Furthermore, the RPE secretes neuroprotective factors, such as the pigment epithelium-derived factor (PEDF), that helps to maintain retinal homeostasis (Barnstable and Tombran-Tink, 2004; Strauss, 2005). The RPE is also a key element in the physiopathology of the retinal detachment. In this sense, our data obtained from the coculture of RPE cells with neuroretina explants, suggested a neuroprotective role of the RPE (Di Lauro et al., 2016), that can be linked to the neurotrophic factors secreted or induced by RPE cells during coculture. Thus, in this study we quantified the PDEF levels in a coculture model of physically separated neuroretina and RPE, and evaluated its importance for neuroretinal maintenance.

Methods

Experimental samples obtention

Porcine eyes were obtained from the local slaughterhouse and transported to the laboratory in ice-cold Dulbecco's Modified Eagle Medium (DMEM) supplemented with a 10% antibiotic-antimycotic (both Gibco, Invitrogen, Paisley, UK). In a cell culture laboratory, the eyes were immersed in 70% ethanol during 2 minutes and washed in clean DMEM supplemented with 10% antibiotic-antimycotic.

RPE cell isolation and culture

RPE cells from the porcine eyes were obtained as previously described by our group (Di Lauro et al., 2016; Rodriguez-Crespo et al., 2014). Passage 2 RPE cells were seeded (30,000 cells/cm^2) (Di Lauro et al., 2016; Rodriguez-Crespo et al., 2014) on the bottom of Transwell® culture plates (Corning Inc., Corning, NY) and grown for 24 hours in complete DMEM before co-culturing with neuroretina explants. RPE cell morphology, viability and cell numbers during cultures were evaluated by phase contrast microscopy and trypan blue exclusion assay (Sigma-Aldrich, St. Louis, MO), respectively.

Central neuroretina explant preparation and culture

Neuroretina explants were obtained from porcine eyes as previously described by our group (Di Lauro et al., 2016). The neuroretina explants were laid over the Transwell® membranes (24-mm diameter with 0.4-µm pore polycarbonate membrane insert; Corning Inc.) with the photoreceptor layer facing the membrane. Thus, neuroretina explants were co-cultured with RPE cells (30,000 cells/cm2) but physically separated from them by the culture medium overlying the RPE cells and the Transwell® membrane. Other explants were cultured as controls without RPE cells. Cultures were maintained during 9 days in 1:1 Neurobasal-A/DMEM supplemented with 10% fetal bovine serum, 2% B-27 (Gibco), 1% L- glutamine (Sigma-Aldrich, St. Louis, MO), and 1% antibiotic-antimycotic mixture and maintained in standard culture conditions.

Pigment epithelium-derived factor (PEDF) quantification

The concentration of PEDF was determined in the incubation medium of neuroretinas cultured alone or co-cultured with RPE cells. The media were collected at medium exchange on culture days 3, 5, 7, and 9 (n=5 each), and stored at -80ºC until assayed. The co-culture medium was analyzed as a control for nonspecific binding. PEDF levels were measured with quantitative enzyme-linked immunosorbent assays (ELISA) specific for porcine PEDF (Pig PEDF ELISA Kit; My Bio Source, San Diego, CA). The mean minimum concentration of porcine PEDF detected with this method was 0.025 ng/ml, according to the manufacturer. The detection range of this ELISA was 0.05-10 ng/ml. Plates were read in a SpectraMAXRM5 multidetection microplate reader (Molecular Devices, Sunnyvale, CA) at 450 nm (reference wavelength: 540 nm).

Statistics

Statistical analysis was performed using R Statistical Software version 3.1.0 (Foundation for Statistical Computing, Vienna, Austria). Pair-wise Student's-tests were used to compare mean PEDF concentrations on days 3, 5, 7, and 9 of neuroretina culture alone and in co-culture with RPE cells. The statistical significance level was set at 5%. Data were expressed as means ± SEM.

Results

Neuroretina and RPE cultures and cocultures

During neuroretinal explants culture and neuroretinal explants with RPE cocultures, the neuroretinal parenchyma and the RPE cells showed similar morphological results (data not shown) to those previously described by our group (Di Lauro et al., 2016).

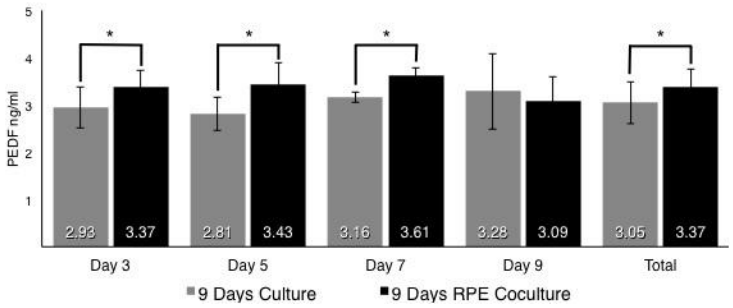


Figure 1. Pigment epithelium-derived factor (PEDF) levels (ng/ml) during culture and co-culture period The total average PEDF concentration of cultures without RPE was significantly lower than that of cultures with RPE. On culture days 3, 5, and 7, PEDF concentrations were higher in media collected from neuroretinas co-cultured with RPE than alone. On day 9 of culture, the difference between them was not significant. *p<0.05.

Pigment epithelium-derived factor (PEDF) quantification

The incubation media collected from neuroretina cultures without and with RPE cells contained detectable levels of PEDF at both conditions and all time-points evaluated (Fig. 1). For culture days 3, 5, and 7, and through the whole time of culture, the PEDF concentration was significantly higher for neuroretinas cultured with RPE ($p<0.05$). For the 9 days of culture, the average PEDF concentration for cultures without RPE (n=5), 3.04±0.44 ng/ml, was lower than that for cultures with RPE (n=5), 3.37±0.37 ng/ml, but the difference was not significant (Fig. 1).

Discussion

The physical separation between the neuroretina and the RPE, such as in retinal detachment, starts a cascade of events that result in cellular changes throughout the retina (Pastor et al., 2002). These events are in part responsible for the poor functional results that occur in some patients after successful reattachment surgery (Fisher et al., 1995). Neuroretina cultures represent a useful tool to study retinal modifications in vitro, because they closely mimic cellular and molecular dynamics in vivo, and due to their easy development and relatively inexpensiveness. However, the RPE also plays an important role in the retinal degeneration process of the retinal detachment (19). Thus, RPE cells are necessary for the viability and functionality of the retinal layers, as these cells secrete growth and trophic factors that act on the neuroretina (10). To overcome the absence of RPE cells in neuroretinal cultures, our group previously developed and characterized a novel organotypic co-culture system in which the porcine neuroretina was co-cultured with RPE cells. In this coculture model it was prevented any physical contact between the co- cultured cells but allowing signal molecules to pass between them (Di Lauro et al., 2016). Based on morphological and immunohistochemical results found in our mentioned study (Di Lauro et al., 2016), neuroretina preservation in co-cultures with RPE cells may be linked to neuroprotective factors secreted by RPE cells. In this study, we determined the PEDF levels, a RPE-secreted neuroprotective factor, in the co-culture model of porcine neuroretina and RPE cells, as a possible molecule implicated in the RPE-neuroretina neuroprotection in our ex vivo system.

PEDF is a neurotrophic/neuroprotective glycoprotein secreted by RPE and Müller cells with great importance for retinal survival and functionality (Barnstable and Tombran-Tink, 2004). This molecule has activities associated with retinal neuroprotection, neurite outgrowth, and photoreceptor morphogenesis (Jablonski et al., 2000). Furthermore, it is able to protect the innermost retina and ganglion cells from death induced by ischemia or cytotoxic agents (Pang et al., 2007). In vivo, PEDF is secreted across the RPE apical surface where it acts on neurons and photoreceptors. It is thought to be one of the most important neuroprotective agents secreted by the RPE, providing protection for photoreceptors against glutamate-, hypoxia-, light- and genetically-induced apoptosis and to act as an antiangiogenic factor (Strauss, 2005). PEDF has an important antiapoptotic effect mainly due to the induction of Bcl-2 expression to prevent nuclear translocation of apoptosis-inducing factor (Murakami et al., 2008). The overexpression of Bcl-

2 prevent Bax translocation/activation and result in lower cell apoptosis (Ku et al., 2011). It is for that reason that we selected it as a biomarker of RPE-neuroretinal interaction in our *ex vivo* system. Furthermore, the secretion of neuroprotective factors by RPE cells, such as PEDF and ciliary neurotrophic factor, is stimulated during retinal damage (Strauss, 2005).

In the experiments presented here, the PEDF concentration was significantly higher in the co-cultures than in the isolated neuroretina cultures. However, previous studies (Jablonski et al., 2000; Pang et al., 2007), showed that PEDF concentration secreted by early passage RPE cells culture alone was considerably higher (pg/ml) than the levels detected by us (ng/ml). Neuroretina explants are also able to secrete PEDF spontaneously (Unterlauft et al., 2014), and the basal PEDF levels shown by the neuroretinas cultured alone in our experiments may be attributed to secretion by Müller cells, as described after retinal damage and hypoxia (Lange et al., 2008). However, it was not within the scope of these experiments to identify the cellular source of PEDF or to determine the PEDF levels in RPE cultures alone or in direct contact with the neuroretina. Furthermore, the sample size used to analyze PEDF concentration (n=5) was limited.

Conclusions

In a co-culture model of porcine neuroretina supplemented with RPE cells, maintained separately within the same culture wells, the RPE cells neuroprotective role may be linked to the beneficial effects of neurotrophic factors, such as PEDF, secreted or induced by RPE cells during co-culture. Further studies will help us to clarify if the high PEDF levels detected in the culture media are directly related to the retinal preservation in the RPE co-cultures.

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